

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 25

1

Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.5, 16.6 (curl, divergence, parametric surfaces)
- Exam 3: Friday May 1 – Ch 13.1-13.3 and 16.1-16.8
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

2



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3

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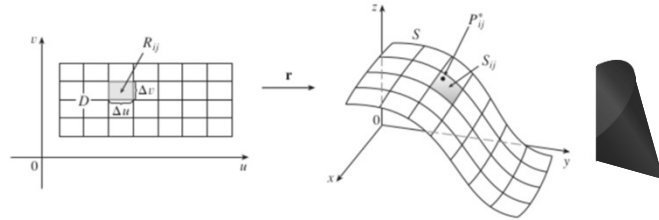
Surface Integrals



4

Integrals on Parametrized Surfaces

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$



$$\iint_S f(x, y, z) \, dS = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(P_{ij}^*) \, \Delta S_{ij}$$

$$\iint_S f(x, y, z) \, dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

5

Integrals on Graphs of Functions

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

$$\iint_S f(x, y, z) \, dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

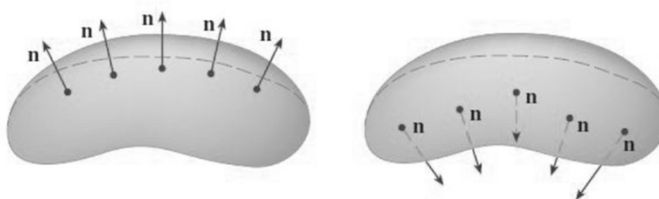
$$\iint_S f(x, y, z) \, dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \, dA$$

6

Orientation

- For surface integrals of a vector field $\vec{F}$ over a surface S , the orientation of the surface must be specified, just like the line integrals. The orientation of a surface is specified via its normal vectors at any point.
- For a closed surface S , we define **the positive orientation** of S to mean that all normal vectors to S point outward (with negative meaning normal vectors point inward instead). If it is impossible to have all normal vectors always point either outward or inward, then the surface S is non-orientable.

FIGURE 7
The two orientations
of an orientable surface



7

- To define a surface integral of a vector field, we only want to count the portion of $\vec{F}$ that is normal to the surface S at any point. Hence, the dot product again plays a role in the computation.
- Then the **surface integral** of $\vec{F}$ over the surface S is defined as

$$\begin{aligned}
 \iint_S \vec{F} \cdot d\vec{S} &= \iint_S \vec{F} \cdot \vec{n} \, dS \\
 &= \iint_S \vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \, dS \\
 &= \iint_D \left[\vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \right] |\vec{r}_u \times \vec{r}_v| \, dA \\
 &= \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) \, dA.
 \end{aligned}$$

This is often called the **flux integral** as it measures the flux of $\vec{F}$ across the surface S .

- Recall and compare it with the line integral of vector fields:

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt.$$

Orientation

8

Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$. (Assume S is positively oriented.)

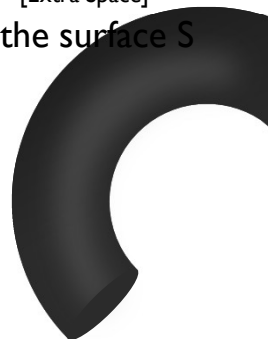
9

Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$. (Assume S is positively oriented.)

[Extra Space]

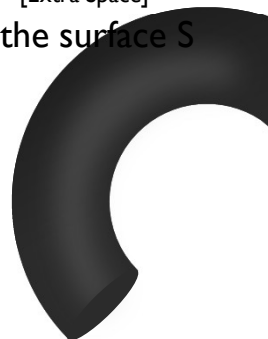
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[Extra Space]
 Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$. (Assume S is positively oriented.)



11

[Extra Space]
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12

Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \text{curl}(F) \cdot dS$$

13

Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \text{curl}(F) \cdot dS$$

14

[Extra Space]

Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \text{curl}(F) \cdot dS$$

15

Questions?

16



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17

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Stokes' Theorem



18

Recall: Green's Theorem

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\int_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

19

Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P \, dx + Q \, dy \quad (\text{curl } \mathbf{F}) \cdot \mathbf{k} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \cdot \mathbf{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Then Green's Theorem says:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_D (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dA$$

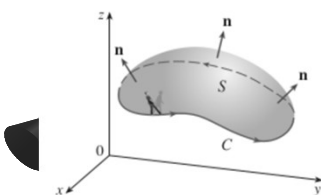
20

Stokes' Theorem

Stokes' Theorem

Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S . Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$$



Sir
George Stokes
FRS



Stokes, 1860s

35th President of the Royal Society

In office
1885–1890

Preceded by Thomas Huxley

Succeeded by Lord Kelvin

Personal details

Born George Gabriel Stokes
13 August 1819
Skreen, Ireland, United Kingdom of Great Britain and Ireland

Died 1 February 1903 (aged 83)
Cambridge, England, United Kingdom of Great Britain and Ireland

21

Example: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = (-y^2, x, z^2)$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. (C is oriented counterclockwise when viewed from above.)

22

[Extra Space]

Example: Evaluate $\int_C F \cdot dr$ where $F = (-y^2, x, z^2)$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. (C is oriented counterclockwise when viewed from above.)

23

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

24

[Extra Space]

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

25

[Extra Space]

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

Alternative solution: Change S to a disk!

26

[Extra Space]

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

Alternative solution: Change S to a disk!

27

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the hemisphere $x^2 + y^2 + z^2 = 9, z \geq 0$ and oriented upwards.

28

[Extra Space]

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the semisphere $x^2 + y^2 + z^2 = 9$, $z \geq 0$ and oriented upwards.

29

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 4$ along with the top at $z = 4$.

30

[Extra Space]

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 4$ along with the top at $z = 4$.

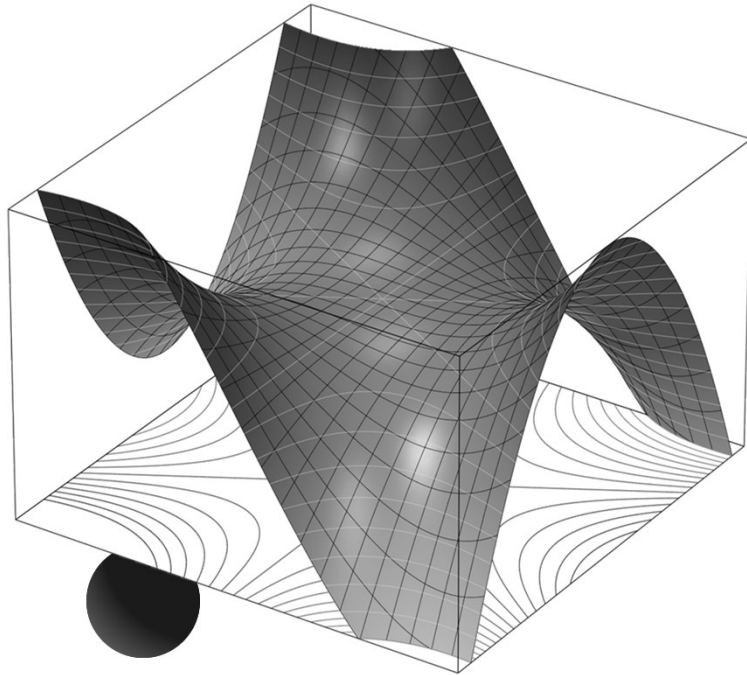
31

Questions?

32

Thank you

Until next time.



33

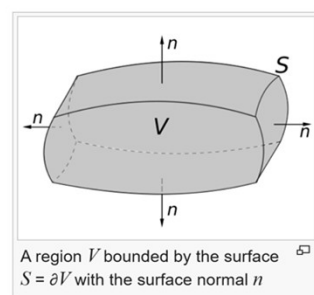
The Divergence Theorem (Gauss's Theorem)

Mathematical statement [edit]

Suppose V is a subset of $\mathbb{R}^n$ (in the case of $n = 3$, V represents a volume in three-dimensional space) which is compact and has a piecewise smooth boundary S (also indicated with $\partial V = S$). If $\mathbf{F}$ is a continuously differentiable vector field defined on a neighborhood of V , then:^{[4][5]}

$$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \oint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) \, dS.$$

The left side is a volume integral over the volume V , and the right side is the surface integral over the boundary of the volume V . The closed, measurable set ∂V is oriented by outward-pointing normals, and $\hat{\mathbf{n}}$ is the outward pointing unit normal at almost each point on the boundary ∂V . ($d\mathbf{S}$ may be used as a shorthand for $\mathbf{n}dS$.) In terms of the intuitive description above, the left-hand side of the equation represents the total of the sources in the volume V , and the right-hand side represents the total flow across the boundary S .



34